

SHEMYA ISLAND, AK, USA

WMO: 704140

Lat: 52.717N

Lon: 174.100E

Elev: 37

StdP: 100.88

Time Zone: -10.00 (NAH)

Period: 94-19

WBAN: 45715

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-4.9	-15.1	1.0	-1.8	-12.7	1.3	-2.8	23.9	2.3	22.2	2.2	7.0	300	1.008

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.4	13.1	10.9	12.3	10.6	11.7	10.4	11.8	12.2	11.3	11.7	10.8	11.2	7.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	8.4	11.7	11.0	8.2	11.4	10.5	7.9	11.0	33.6	12.3	32.4	11.6	31.5	11.3	14.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.0	16.0	DB	-7.9	14.3	1.0	1.4	-8.6	15.2	-9.2	16.0	-9.8	16.8	-10.5	17.8	(4)
(5)			WB	-9.0	13.0	1.2	1.0	-9.8	13.7	-10.5	14.3	-11.2	14.9	-12.1	15.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-0.3	-0.1	0.3	1.8	3.9	6.1	8.9	10.4	9.7	6.1	3.1	0.4		
	DBStd	4.18	2.37	2.00	2.09	1.30	1.06	1.06	1.48	1.14	1.29	1.90	1.82	2.39		
	HDD10.0	2154	321	283	302	246	188	117	42	9	20	121	208	299		
	HDD18.3	5156	579	517	560	496	446	367	293	247	259	378	458	557		
	CDD10.0	39	0	0	0	0	0	0	7	21	11	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	8.1	9.4	9.7	9.1	8.5	7.2	6.5	5.5	6.4	7.6	8.7	9.6	9.5		
(15) Precipitation	PrecAvg	804	62	49	47	45	42	43	69	97	75	97	100	75		
	PrecMax	1070	143	87	90	101	77	79	221	238	128	218	203	152		
	PrecMin	439	31	18	11	9	19	4	18	19	43	48	25	26		
	PrecStd	144	28	19	21	24	14	20	39	55	24	38	42	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.4	5.1	5.3	7.5	9.7	16.2	14.1	13.3	11.1	8.2	6.4		
		MCWB	3.3	4.0	3.9	3.4	5.1	7.4	9.6	11.9	10.9	9.7	7.2	5.7		
	2%	DB	4.0	3.5	4.3	4.6	6.5	8.7	13.6	13.0	12.4	10.1	7.1	5.1		
		MCWB	2.9	2.9	3.1	3.0	4.5	7.0	9.0	11.3	10.5	8.7	5.9	4.6		
	5%	DB	3.5	3.0	3.8	4.0	6.0	8.0	11.5	12.4	11.9	9.5	6.3	4.3		
		MCWB	2.5	2.1	2.7	2.7	4.3	6.7	9.5	11.0	10.3	8.1	5.3	3.4		
	10%	DB	3.0	2.5	3.2	3.6	5.4	7.5	10.7	11.9	11.4	8.9	5.8	3.5		
		MCWB	1.9	1.5	2.1	2.4	4.1	6.5	9.5	10.7	10.0	7.6	4.7	2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.1	4.5	4.2	6.1	8.7	11.5	12.8	12.0	10.3	7.8	5.8		
		MCDB	4.4	4.2	4.7	4.5	6.7	9.2	12.3	13.4	12.3	10.4	8.0	6.0		
	2%	WB	3.4	3.2	3.6	3.6	5.3	7.9	10.8	12.0	11.4	9.3	6.5	4.8		
		MCDB	3.8	3.4	4.0	4.0	5.7	8.2	11.2	12.4	11.8	9.7	6.8	5.2		
	5%	WB	2.8	2.5	3.0	3.3	4.9	7.4	10.3	11.6	11.0	8.6	5.7	3.7		
		MCDB	3.3	2.8	3.6	3.7	5.4	7.7	10.7	12.0	11.4	9.3	6.1	4.2		
	10%	WB	2.1	1.8	2.3	2.8	4.5	6.9	9.8	11.2	10.6	7.9	5.1	2.7		
		MCDB	2.8	2.3	3.0	3.4	5.0	7.3	10.2	11.5	11.0	8.7	5.6	3.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.8	2.8	3.3	3.0	2.8	2.4	3.0	2.4	2.8	3.3	3.2	3.1		
		MCDBR	2.5	3.0	3.2	3.3	3.3	3.0	4.9	3.1	3.2	3.5	3.7	3.2		
	5% WB	MCWBR	2.8	3.3	3.3	2.9	2.6	2.8	3.6	3.1	3.0	3.2	4.1	3.7		
		MCWBR	2.4	2.9	3.0	2.9	2.7	2.6	3.6	3.0	2.7	2.9	3.7	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.370	0.407	0.438	0.446	0.424	0.427	0.411	0.376	0.379	0.374	0.348		
	taud		2.129	2.144	2.091	2.044	2.081	2.235	2.277	2.326	2.377	2.195	2.086	2.102		
	Ebn at Noon		585	707	760	789	805	830	819	813	798	686	542	505		
	Edh at Noon		75	98	126	150	152	132	124	112	95	92	77	65		
(44) All-Sky Solar Radiation	RadAvg		0.67	1.31	2.44	3.40	3.75	3.59	3.57	3.21	2.75	1.83	0.82	0.51		
	RadStd		0.06	0.10	0.23	0.35	0.30	0.30	0.20	0.31	0.25	0.12	0.08	0.05		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page